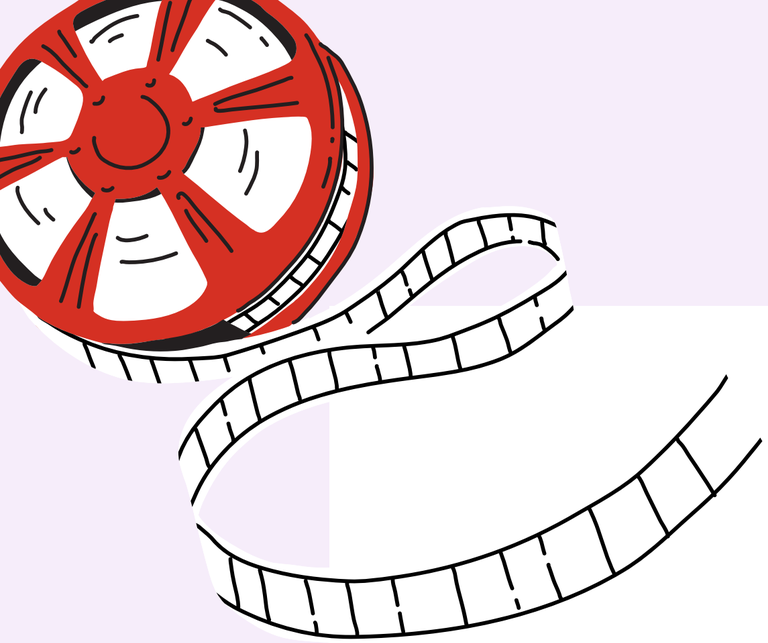






APAN5205

Movie Recommender

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Meet the Team



Sanjana Sankar



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Introduction



The rise of online video streaming platforms has created unlimited access to on-demand content. Unfortunately, this has also led to "choice overload".

Key Challenge: How do users navigate through an overwhelming number of options to find relevant content?

Solution: Content-based recommendation systems.

These systems use inherent features of movies, such as genres, actors, and plot descriptions, to identify similarities between films and produce recommended movies.

Project Overview

We will create & compare 2 recomender models:

1. Genre-based recommender system

2. Hybrid: genre + movie synopsis (NLP) recommender system

Which model will be better? →



Research Question



Does the hybrid (genre + synopsis) recommendation system perform better the genre-based recommendation system?

Null hypothesis: There is no significant difference in performance between the hybrid recommendation system and genre-based recommendation system.

Alternative hypothesis: The hybrid recommendation system perform better than genre-based recommendation system.

Alpha value: 0.1



Our Client: hulu

A quick online search shows the company has a history of customer complaints about its recommendation algorithm—especially low accuracy and a tendency to prioritize its own original content over user preferences.

<u>Business Proposal:</u>
The incorporation of NLP techniques into the recommender system should improve accuracy, reduce choice overload, and increase user engagement.



Our Data





Data Sources

2 IMDb datasets, contains

1) titles and release year
2) ratings





1 Kaggle dataset that contains plot summaries of movies.

Workflow

Data Cleaning

- Standard data pre-processing (reordering, imputations, removing rows/columns)
- Improved reliability of movie ratings with Bayseian weighted rating

Creating Models

- Created genre-only model
  - Genre Vectorization
  - Cosine Similarity Matrix
  - Duplicated Titles
- Created hybrid model
  - 3 attempts
  - Final hybrid model relies on 80% plot and 20% genre

Evaluation

- Paired t-test
- Shanon Entropy test: measures the average uncertainty, surprise, or information content in a system's outcomes

Live Demo





Deploy

NLP app

genre app

Genre-Based Movie Recommender

Type a movie title to get similar movies based on genre similarity.

Search movie title

Number of recommendations

20

5 20

Recommend

Evaluation

Build the test set

13 test movies
Set scoreboard for genre model & hybrid model

Run the test for both models

Score = average of the Weighted Rating
of recommended movies

Paired T-test

Test whether the recommended movie quality difference between Genre and Hybrid models is meaningful, not random

P-Value

- p-value < alpha (0.1) : Significant
- P-value ⩾ alpha (0.1) : Inconclusive

Result

p-value = 0.0308

Shannon Entropy
Genre Model = 1.4090
Entropy = 2.6806

Findings: So which model did best?

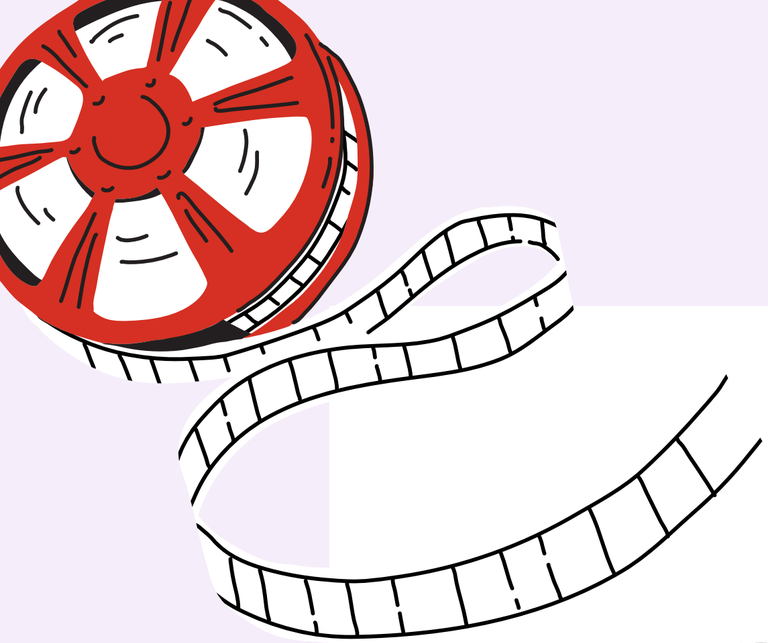
NLP Hybrid Model

Findings: So which model did best?

Hybrid model did better across both evaluations.

We saw a p value of 0.0308 in the paired t-test and a higher genre diversity value with the Shanon Entropy test (2.6806 in comparison to 1.4090 for the genre based model).

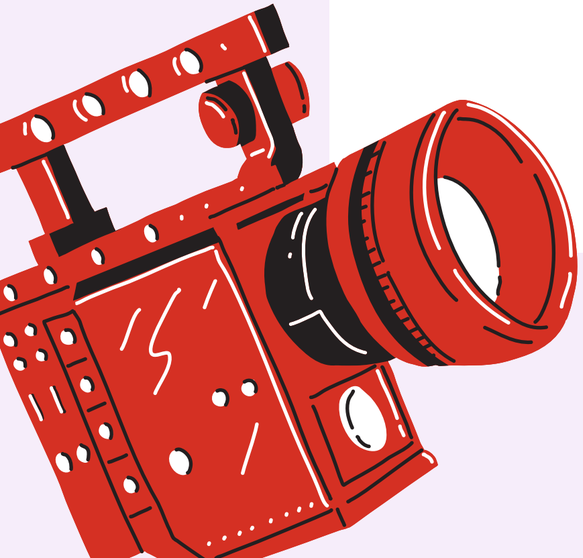
For the future, it would be useful to add in more features such as actors and directors to further broaden the creativity of the suggestions.







Questions?







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